

Micah Scheaffer

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Objective Statement

Mechanical and cyber-physical systems engineering student passionate about robotics, emerging technology, and leadership. Seeking a full time role in robotics or mechanical design.

Technical Skills

CAD/CAM: Inventor, Fusion 360, Creo, Solidworks; CNC Machining, 3D-Printing (Bambu, UltiMaker, Crealty)

Programming Languages/Tools: Python, MATLAB, Simulink, Linux VMs, Git, Claude Code, Node-RED, Google Apps Script

Robotics & AI: ROS, OpenCV, YOLO, PID Control, Odometry

Office & Productivity: LLM Prompting, Microsoft Office Suite, Google Workspace, Notion

Other: Finite Element Analysis (Creo Simulate), Failure Mode and Effects Analysis, Systems Integration, Technical Writing

Education

Undergraduate Student | Iowa State University | Ames, IA

August 2023 - December 2026

- Majoring in Mechanical Engineering with a minor in Cyber-Physical Systems; 3.63 GPA, 4x Dean's List
- Coursework Highlights: Mechanical Considerations in Robotics (graduate level), System Dynamics and Controls, Design and Analysis of Cyber-Physical Systems, Fundamentals of Cyber-Physical Systems

Experience

Incoming Automation Controls Intern | Molex | Conway, AR

May 2026 - August 2026

Engineering Design Intern | Caterpillar, Inc. | Decatur, IL

May 2025 - July 2025

- Redesigned platforms and handrails for 30% weight and cost reduction by reducing welds and commonizing parts
- Validated structural integrity of improved designs using FEA in compliance with ISO and MSHA safety standards
- Improved product safety with a case study on standardizing fire extinguisher placement for off-highway trucks

Engineering Design Parallel Co-op | Superior Industries | Columbus, NE; Remote

May 2024 - January 2025

- Modeled a parametric template to derive 75+ assemblies, parts, and prints of various sizes
- Created a standardized parts library for hundreds of HDPE pipe components
- Optimized calculation and data organization using spreadsheets and Google Apps Script

Leadership

President, Founder & Systems Lead | VEX U Cyber Robotics Club

February 2024 - Present

- Established and solicited funding for a competitive robotics club at Iowa State with over 20 members
- Lead a team in the research, design, prototyping, manufacturing, testing, and documentation of multiple robots
- Integrate software and hardware systems using an array of sensors
- Manage a supply chain with considerations in manufacturing capabilities, leadtime, and tariffs

Volunteer Judge | Robotics Education Competition Foundation

August 2023 - Present

- Evaluate robotics teams on design, strategy, and execution at local, regional, and national competitions
- Provide constructive feedback to participants, fostering an educational yet competitive environment

Resident Assistant, RA Advocacy Team Secretary | Dept. of Residence | Ames, IA

January 2025 - Present

- Cultivate a welcoming community for over 50 residents
- Ally with peers and university management to improve RA benefits and resources
- Collaborate with residence directors to interview a competitive pool of RA candidates

Projects

Autonomous Photogrammetry Drone | Personal Project

Ongoing

- Designed, modeled, and constructed an 8" quadcopter with onboard compute, sensors, and custom battery
- Building software pipeline for SLAM navigation and 3D render output using open source tools

Model Smarthome HVAC | Fundamentals of Cyber-physical Systems

Spring 2025

- Developed software for an HVAC testbed with multiple MCUs, sensors, servos, and a thermal unit
- Programmed an automated control algorithm, user interface, and network communications

Awards

Engineering Student Leadership Development Scholarship Recipient

Fall 2024

Vex Robotics World Champion

April 2023

US Open Robotics Championship - Skills Champion & Grand Finalist

March 2023